

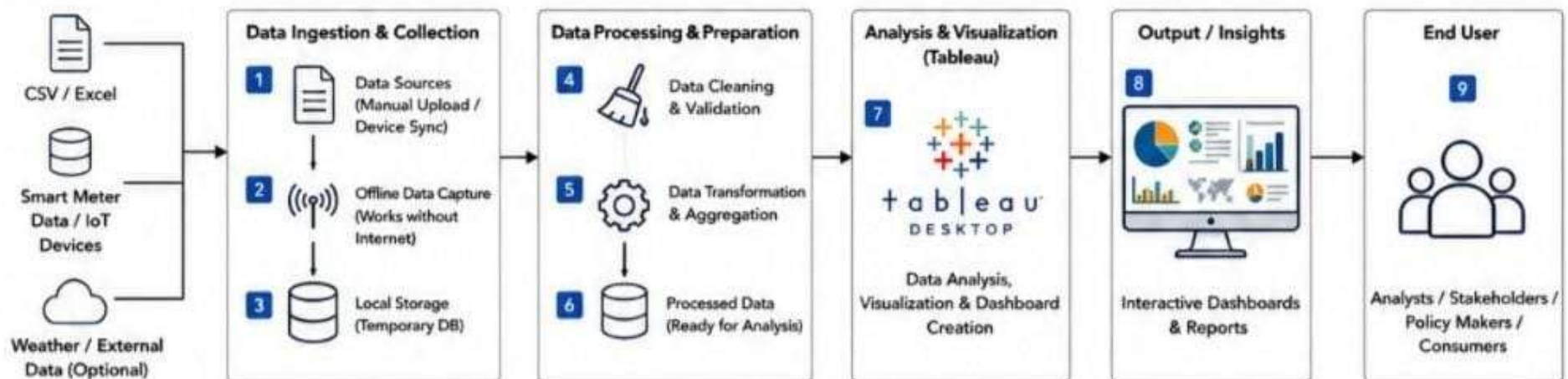
## Project Design Phase-II Technology Stack (Architecture & Stack)

|               |                                                                              |
|---------------|------------------------------------------------------------------------------|
| Date          | 29 July 2026                                                                 |
| Team ID       |                                                                              |
| Project Name  | Plugging into the Future: An Exploration of Electricity Consumption Patterns |
| Maximum Marks | 4 Marks                                                                      |

### Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

### Example:



**Table-1 : Components & Technologies:**

| <b>S.No</b> | <b>Component</b>                     | <b>Description</b>                                                                        | <b>Technology</b>                                     |
|-------------|--------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------|
| <b>1</b>    | <b>User Interface</b>                | <b>Interactive dashboard for visualizing electricity consumption trends and analysis.</b> | <b>Tableau Public / Tableau Desktop</b>               |
| <b>2</b>    | <b>Data Source</b>                   | <b>Stores electricity consumption data for analysis.</b>                                  | <b>CSV Dataset / Excel Dataset</b>                    |
| <b>3</b>    | <b>Data Processing</b>               | <b>Cleans, filters, and prepares electricity consumption data before visualization.</b>   | <b>Tableau Prep / Tableau Desktop</b>                 |
| <b>4</b>    | <b>Data Visualization</b>            | <b>Creates charts, graphs, maps, KPIs, and dashboards for analysis.</b>                   | <b>Tableau</b>                                        |
| <b>5</b>    | <b>Dashboard Filters</b>             | <b>Enables filtering by state, month, year, and consumption category.</b>                 | <b>Tableau Interactive Filters</b>                    |
| <b>6</b>    | <b>Data Analytics</b>                | <b>Performs trend analysis, state-wise comparison, and consumption pattern analysis.</b>  | <b>Tableau Calculated Fields &amp; Analytics</b>      |
| <b>7</b>    | <b>Reporting**</b>                   | <b>Generates reports and dashboard summaries for stakeholders.</b>                        | <b>Tableau Dashboard / PDF Export</b>                 |
| <b>8</b>    | <b>File Storage</b>                  | <b>Stores the dataset and Tableau workbook.</b>                                           | <b>Local File System (.csv, .xlsx, .twb, .twbx)</b>   |
| <b>9</b>    | <b>Project Website (Optional)</b>    | <b>Hosts the Tableau dashboard for online access.</b>                                     | <b>HTML, CSS, JavaScript, GitHub Pages</b>            |
| <b>10</b>   | <b>Dashboard Publishing</b>          | <b>Publishes the dashboard for public viewing and sharing.</b>                            | <b>Tableau Public</b>                                 |
| <b>11</b>   | <b>Infrastructure (Deployment)**</b> | <b>Deployment of the dashboard on local system or cloud platform.</b>                     | <b>Local Computer / Tableau Public / GitHub Pages</b> |

**Table-2: Application Characteristics:**

| <b>Characteristics</b>          | <b>Description</b>                                                                                                                                  | <b>Technology</b>                                            |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| <b>Open-Source Frameworks</b>   | Open-source technologies used for developing and hosting the project website and dashboard.                                                         | HTML5, CSS3, JavaScript, GitHub Pages                        |
| <b>Security Implementations</b> | Access to the Tableau workbook and dataset is restricted to authorized users during development. Published dashboards use secure HTTPS connections. | HTTPS, Tableau Public Security, GitHub Security              |
| <b>Scalable Architecture</b>    | The dashboard is designed to accommodate larger electricity consumption datasets with minimal modifications and supports easy updates.              | Tableau Desktop, Tableau Public                              |
| <b>Availability</b>             | The dashboard is available online through Tableau Public and can be accessed anytime using a web browser.                                           | Tableau Public, GitHub Pages                                 |
| <b>Performance</b>              | Optimized dashboard with interactive filters, calculated fields, and efficient visualizations to ensure fast loading and smooth user interaction.   | Tableau Optimization, Interactive Filters, Calculated Fields |